

VRAI Faces — Research Brief RB-001

Real ARKit-52 Blendshape Rig for the MediaPipe 468-Vertex Topology

ID: RB-001 **Created:** 2026-05-29 **Status:** Open — not started **Gates:** ROADMAP Phase 1.2
Module: mesh_builder (morph_basis) **Relates to:** ADR-0019/0020, ADR-0001, ADR-0005, ADR-0014
Run in: Claude Cowork (deep research)

A self-contained brief defining the parameters for a deferred, gated enhancement. It is intended to be executed later in Claude Cowork; its output feeds an ADR and a drop-in implementation. Nothing here is required to ship the current build — the avatar already renders and exports with a procedural approximation.

1. Objective

Determine the best way to obtain a real, production-quality **ARKit-52 blendshape deformation basis** mapped onto MediaPipe's canonical **468-vertex** face topology, to replace the current procedural approximation (4 of 52 shapes) in `mesh_builder/impl/morph_basis.ts`. "Best" = highest expression fidelity that stays within the project's local-first, deterministic, PHI-safe, clinical constraints.

2. Why this is gated (not done inline)

- No clean off-the-shelf ARKit-52 → MediaPipe-468 deformation basis is known to exist.
- Candidate sources involve **licensing review** and/or a heavier modeling pipeline (e.g. deformation transfer); some may be paid or otherwise gated.
- It must not compromise local-first delivery (ADR-0001), deterministic/auditable output (ADR-0005), the PHI posture (ADR-0014), or clinical fidelity (pain, drowsiness).

3. Current state (context for the researcher)

- **mesh_builder** builds the real 468-vertex mesh from MediaPipe FaceLandmarker landmarks; the live path is browser-verified.
- **Morph deltas today** are a procedural, geometry-driven approximation: `jawOpen`, `mouthSmileLeft/Right`, `browInnerUp` — the other 48 ARKit shapes are zero.
- **avatar_exporter** already bakes whatever deltas exist into glTF `targets[]` by name. So a real rig is a **drop-in**: replace the basis and the export + animation paths already consume it — no contract changes.

4. Research questions

1. Which ARKit-52 rigs / parametric face models exist (e.g. ICT-FaceKit, FLAME, Basel BFM, MetaHuman, Ready Player Me, ARKit reference), and what are their **licenses** for clinical/commercial use *and redistribution in a bundled app*?
2. Which expose ARKit-52 (or cleanly mappable) blendshapes, and how do they map onto the MediaPipe 468 topology (vertex correspondence / deformation transfer)?
3. What is the effort + toolchain for the mapping — a one-time offline bake vs. runtime?
4. Clinical fidelity: do candidates cover **pain, drowsiness**, and other clinically salient affects, or only the standard 52?
5. Bundle-size / local-first impact (delta-data size on disk)?
6. Which options are gated/paid, and on what terms?

5. Evaluation criteria

Criterion	Weight	Notes
License — clinical use + redistributable	MUST-PASS	Bundled, on-device, possibly commercial. Disqualifies non-redistributable assets.
Topology fit / mapping effort	High	Correspondence to MediaPipe 468; one-time bake preferred.
Expression fidelity (incl. clinical affects)	High	Pain / drowsiness coverage is a differentiator for a patient sim.
Bundle size / local-first compatibility	Medium	Delta data served from app origin (ADR-0001), never a runtime CDN.
Determinism & PHI safety	MUST-PASS	Deterministic output (ADR-0005); no facial-image gathering; nothing leaves device (ADR-0014).
Maintenance / provenance	Medium	Stable source, clear versioning, low upkeep.

6. Deliverables (what Cowork should return)

- A ranked comparison of viable options against the criteria above.
- A recommended option + a concrete integration path (e.g. a one-time bake → `face_mesh_morphbasis.json`, loaded like the topology asset is today).
- A license summary and any procurement that would be required.
- A clear **go / no-go** with a rough effort estimate — ready to convert into an ADR.

7. How the result re-enters the build

Research output → a new ADR (next number in `Memory_management.MD §7`) → implementation: produce the baked delta asset, swap `morph_basis.ts` to load it (falling back to the procedural basis if the asset is absent), regenerate, and verify. `avatar_exporter` already bakes the deltas by name, so no contract changes are needed.

8. Hard constraints

- **Local-first (ADR-0001):** assets served from the app origin; no runtime CDN.
- **Deterministic & auditable (ADR-0005):** JSON-defined, reproducible.
- **PHI (ADR-0014):** nothing leaves the device; never gather or scrape facial images.
- **Naming:** ARKit-52 canonical names, matching `mesh_builder` + `avatar_exporter`.

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